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External fixation versus amputation in the treatment of ulcerated diabetic foot: a cost-utility analysis using real-world data and a discrete-event simulation model

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Abstract

Background Diabetic foot is a severe complication of diabetes mellitus, with a significant impact on patients' quality of life. The traditional treatment has been amputation, but its high mortality rate has prompted a search for alternatives, such as the resection of non-viable tissue and limb stabilization using external fixation. This study evaluates the cost-effectiveness relationship between these two strategies.

Methods An assessment was made of the incremental cost-effectiveness ratio (ICER), calculated over a ten-year time horizon, of both amputation and external fixation. The analysis was based on clinical and cost data obtained from a previous study and used a discrete event simulation model.

Results The mean ten-year quality-adjusted life expectancy for amputated patients was 2.641 ± 1.64 quality-adjusted life years (QALYs) as compared with 6.627 ± 1.46 QALYs for patients treated with external fixation ($p < 0.001$). The average cost-effectiveness ratio (CER) was estimated at €29,717/QALY for amputation and €15,360/QALY for external fixation. A comparison between both options resulted in an ICER of €5,847/QALY. Sensitivity analyses found ICER values ranging between €5,847/QALY and €12,296/QALY. A budget impact analysis revealed that the incorporation of the new treatment alternative into the Spanish National Health System would lead to a budget increase of €528.2 million over a ten-year horizon.

Conclusions The use of external fixation in the treatment of ulcerated diabetic foot was shown to be a cost-effective option compared to major amputations.

Keywords Diabetic foot, Foot ulcer, Amputation, External fixator, Ilizarov, Economics, Cost, Quality of life, Charcot, Burden

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Introduction

Septic ulceration of the foot is one of the complications of diabetes mellitus with greatest impact on patients' quality of life [1]. The condition originates from diabetic neuropathy, which causes loss of sensitivity and deformities in the foot, altering its normal biomechanics. The appearance of abnormal pressure points could result in subcutaneous hemorrhages which typically develop into ulcers over time. Loss of sensitivity means that the patient is unable to feel the pain, which delays the initiation of treatment. At the same time, these patients often suffer from peripheral arterial disease, which tends to hamper healing, increasing the risk of gangrene [2]. Until recently, the only treatment available for the most severe cases was amputation [3], a procedure resulting in mortality rates between 41% and 70% [4–7], comparable to those associated with some of the most aggressive cancers [8–13].



Fig. 1 Circular external fixator (TrueLok, Orthofix Srl, Italy) comprising a series of rings that surround the limb and stabilize the bone by means of pins or wires in a minimally invasive way (Image used with permission from Orthofix Srl, Italy)

Diabetic foot also represents a significant economic burden. In 2015, the direct costs associated with the treatment of cancer in the United States were equivalent to those corresponding to the care of diabetic foot [12], which are typically higher than those of other serious conditions such as depression [13–16]. Nonetheless, apart from the direct cost of ulceration, the condition is associated with high indirect costs mainly related to productivity losses and premature deaths [17, 18]. The quantification of the expenses related to diabetic amputations has prompted the production of abundant literature [19–30], albeit with heterogeneous results, given the diversity of methodologies employed and the diverse geographies and timeframes considered in the studies [29, 30].

A new treatment modality has emerged in the last few years, based on resecting the infected bone and reconstructing the foot by means of circular external fixators [10, 31–37] (Fig. 1). This new approach has been shown to reduce the amputation and mortality rate [31, 32, 34, 35, 37]. However, although the results offered by external fixation have proved highly encouraging, cost analyses for the procedure have been scarce [22, 37] with no cost-utility studies having been published to date.

In the current socioeconomic environment, where healthcare authorities are under growing pressure to reduce costs while maintaining or even improving the quality of care [38, 39], it is crucial to be able to resort to economic analyses conducive to well-founded decision making. Nevertheless, no full economic evaluations have to the best of our knowledge been published that compare the two treatment methods discussed in this study.

The purpose of the present analysis is to carry out a comprehensive cost-utility study of amputation vs. external fixation in the treatment of septic diabetic foot. The analysis is based on a previous clinical study that compared the clinical results of both kinds of treatment and calculated the direct costs associated to them.

Materials and methods

Target population

Adult patients of both sexes presenting with diabetes mellitus complicated by (Brodsky type 1 or Eichenholtz stage III) chronic diabetic foot [40] with deep infected ulcers. No other exclusion criteria were applied.

Econometric model

A cost-utility analysis was carried out, where both health outcomes and the cost of the two interventions studied were evaluated. Costs were measured in monetary units, while health outcomes were expressed in quality-adjusted life years (QALYs) [41–45]. The treatment alternatives compared were amputation of the limb with resection of unviable tissue followed by stabilization with a circular external fixator. The study was carried out from

Table 1 Sources of information for the model and sensitivity analysis

Parameter	Source
General model	
Clinical outcome: Quality of life (EQ-5D-3 L index)	López Capdevilla (2024) [37]
Costs, as evaluated by micro-costing	López Capdevilla (2024) [37]
Comparators: Amputation vs. external fixation	López Capdevilla (2024) [37]
Aggregated patient data: age & sex	López Capdevilla (2024) [37]
Survival associated with each treatment alternative: Kaplan-Meier and Weibull	López Capdevilla (2024) [37]
Time-adjusted health outcomes: Normalized values for diabetics in Spain	Mateo (2015) [48]
Life expectancy of diabetic patients in Spain	Tomic (2022) [49]
Effectiveness thresholds considered	Sacristán (2020) [50] y NICE [51]
Budgetary impact analysis	
Calculation formula	Brosa (2005) [52]
Diabetic amputations in Spain	López de Andrés (2022) [53] and López de Andrés (2015) [54]
Evolution of diabetic amputations	López de Andrés (2022) [53]
Costs associated with surgery	López Capdevilla (2024) [37]
Cost of follow-up	López Capdevilla (2024) [37]
Cost discount rate: 3%	Ministry of Health (2023) [55]
Sensitivity analysis	
SA-1: Higher cost of ext. fixator	€20,000 in addition to surgical costs
SA-2: Poorer results with external fixation	5% reduction of QALE
SA-3: Better results with amputation	5% increase of QALE
SA-4: Poorer scenario for external fixation	SA-2, SA-3 and SA-4 combined
SA-5: Constant number of surgeries	The amputation rate does not decrease
SA 6 – 5% discount rate	Ministry of Health (2023) [55]

the perspective of the health system with a time horizon of 10 years.

A discrete-element simulation model was used to determine the number of QALYs gained from each treatment option during the period analyzed [46, 47]. The model was based on data gathered in a previous clinical and cost analysis [37] as well as on several bibliographic sources (Table 1).

Sources of information and assumptions of the model

The study used as a basis for this analysis [56] provides information about the age and sex of every patient as well as on their date of diagnosis, their date of surgery and, when appropriate, their death date. Quality of life progression was evaluated through the EQ-5D-3 L questionnaire [57, 58], scores being determined using the time trade-off method, considered particularly well-suited for

Table 2 Survival rate estimated based on the treatment method applied

Year	Amputation group	External fixation group
1	85.9%	96.3%
2	75.5%	93.3%
3	68.9%	91.1%
4	63.0%	89.1%
5	59.1%	87.7%
6	50.3%	84.3%
7	45.1%	82.1%
8	42.9%	81.0%
9	39.2%	79.1%
10	35.4%	77.2%

economic evaluations [59]. All the costs associated with the treatment of each individual were also recorded, including diagnostic tests, use of antibiotics, walking aids, medical appointments, realignment surgeries, other surgical procedures, hospital stay, amputations, surgical materials and rehabilitation.

The model assumes that full recovery is achieved at one year from surgery, although it does allow for the possibility that patients may pass away at any time after the procedure. The simulation takes into consideration two possible final health outcomes for each individual: survival or death. For the former, quality of life is modeled to decrease at the same pace as it does for the standard population of diabetic patients of the same age and sex [48]. For the latter, the subject's quality of life plummets to zero, the occurrence of death being modeled on the basis of the life expectancy of diabetic patients in Spain [49]. Given that the previous study does not provide data for the life expectancy of diabetic patients beyond the age of 80, a decision was made to use the remaining life expectancy values from another study for our patients aged 75 or over [60], reducing it by 6 years [49]. Moreover, with the passage of time, quality of life must be adjusted according to the survival expected for each treatment option. To do this, the life expectancy calculated in the previous study using Kaplan-Meier curves [37] was extrapolated by means of a Weibull parametric estimation (Table 2). This analysis was conducted using the *{flexsurv}* R software package [61]. The general functioning of the model is summarized in Fig. 2.

Quality-adjusted life expectancy (QALE) is determined by the QALY area under the curve, which makes it possible to compare different treatment options and relate them with their individual costs.

Decision-making criteria

The mean cost-effectiveness ratio (CER) was calculated for each treatment alternative by dividing the mean cost of the intervention by its net result [62, 63]. The economic evaluation of the two treatment options was

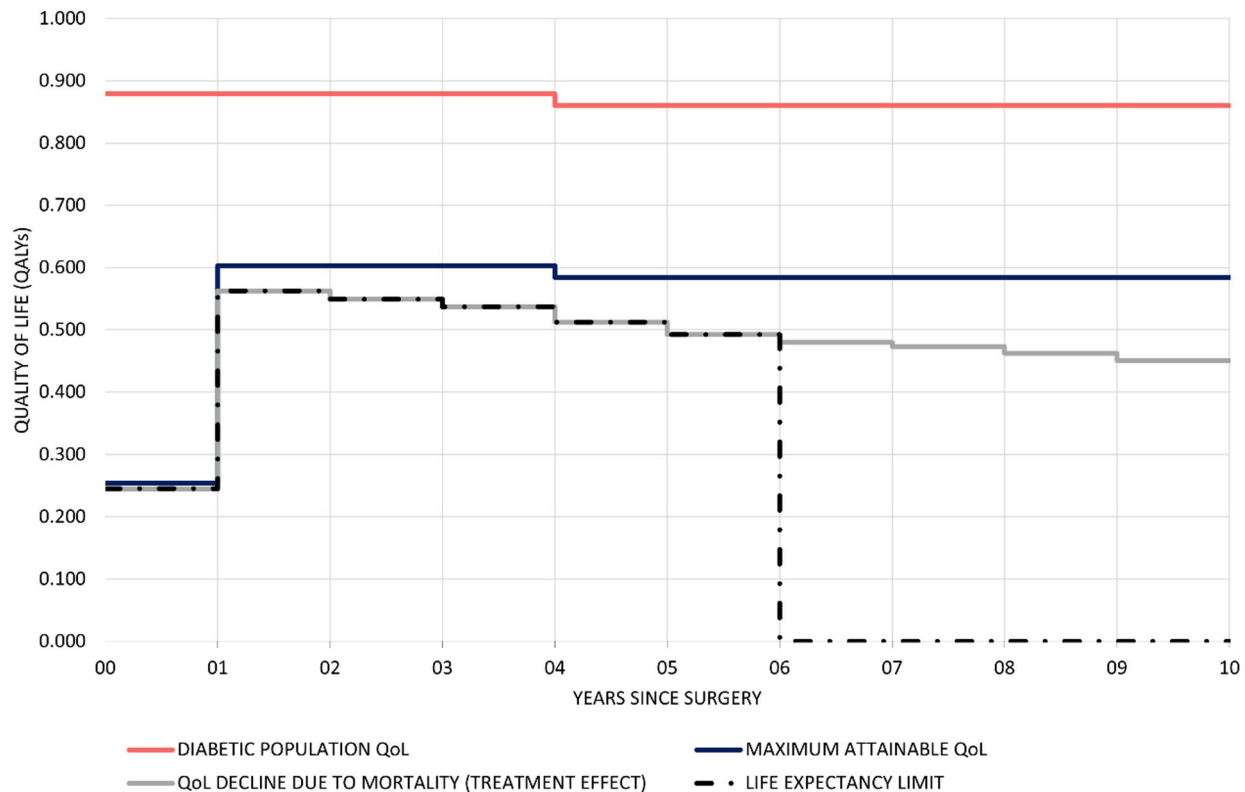


Fig. 2 How the model works for an individual patient. The red line represents the evolution of quality of life over time for a standard diabetic patient of their same age and sex. The blue line indicates the maximum quality of life achievable in their current health condition. During the first year (from year 0 to year 1), this line reflects the improvement in the patient's quality of life; subsequently it decreases at the same pace as among the standard population of diabetic patients of the same age and sex. The gray line shows quality of life reductions, proportional to the death risk associated with the corresponding treatment. The black dotted indicates the point at which quality of life drops to zero, which coincides with the patient's maximum life expectancy. The area below this line represents the quality-adjusted life expectancy, which constitutes the model's final result

carried out based on the incremental cost-effectiveness ratio (ICER). The costs defrayed prior to surgery were not included in the calculation of the ICER, as they were independent from the selected treatment options. To determine whether the use of external fixation was cost-effective, the ICER was compared with the most widely-used reference thresholds [50, 51, 64–67].

Budgetary impact analysis

Assuming that the number of patients treated by means of either treatment option was the same, the budgetary impact analysis should be based on application of the following formula [42, 52]:

$$\text{Budgetary impact} = (MC_{CEF} - MC_{AMP}) \cdot N$$

where MC stands for the mean cost per patient and N for the number of patients being treated. As was the case for the ICER, only incremental costs were considered. The costs corresponding to the surgical procedure were only considered during the year where the operation was performed, whereas the follow-up costs were distributed

over a 2-year period. A patient may be considered dead because they have reached their life expectancy or as a result of applying the mortality rate corresponding to the treatment they received.

Given that the minimum dataset in the Spanish Specialized Care Activity Registry [68] does not distinguish minor from major amputations, we resorted to the literature to estimate the number of patients undergoing a major amputation every year. We found out that between 2017 and 2019 a total of 8,155 major amputations were performed in Spain in patients with type 2 diabetes [53], which entails around 2,718 amputations a year. Given that 95.87% of amputations in Spain are performed in patients with type 2 diabetes [54], we calculated that the total number of annual amputations stands at 2,835, this figure exhibiting an annual decrease of 4% [54].

Discount rates

A cost discount rate of 3% was used for the budgetary impact analysis [43]. As regards health outcomes, as we took into consideration the deterioration of quality of life resulting from aging and the expected death rate

according to the individuals' life expectancy [49] and treatment received [37], no additional time adjustments were considered necessary.

Sensitivity analysis

The sensitivity analysis includes modifications of the parameters affecting the decision-making criteria and of those affecting the budgetary impact.

The following scenarios were considered:

SA-1: higher cost of the external fixator

Given that one of the most significant costs of the treatment is the external fixator itself, in this scenario the price of the fixator was €20,000 higher than the amount paid by the hospital where the patients in the study that served as a basis for this analysis were treated.

SA-2: poorer clinical results with external fixation

The QALYs gained by each patient treated with an external fixator is 5% lower every year than those considered in the study serving as a basis for the present one.

SA-3: better clinical results with amputation

The QALYs gained by patient treated by amputation is 5% higher every year.

SA-4: poorer outcomes with external fixation

This scenario combines scenarios SA-2, SA-3, and SA-4, i.e. it includes an increase in the price of the fixator, a worsening of the results of external fixation, and an improvement in the results of amputation.

SA-5: constant number of surgeries

For the budgetary impact analysis, we considered the possibility that the number of surgeries performed every year would decrease by 4%. This scenario, however, considers the number of surgeries to remain constant throughout the time period analyzed.

SA-6: higher cost discount rate

The cost discount rate is 5% instead of 3%.

Statistical analysis

A descriptive analysis of the data was performed, with central tendency and dispersion measures. The change in quality of life between pre-op and post-op was evaluated using repeated measures contrasts. The differences between the groups were assessed by Student's t test or Mann-Whitney's U test, depending on the distribution of each variable. Statistical significance was set at a p value of 0.05 in all cases. The analysis of data was carried out using R software (R Development Core Team), 4.4.2 [69].

Results

Description of the cohort in the study serving as a basis for this analysis

The previous study included 31 patients (24 males and 7 females). Eighteen (58.1%) were treated by major amputation and 13 (41.9%) with external fixation. Mean follow-up from diagnosis was 63.2 ± 35.2 months, the patients' mean age at the time of treatment was 64.1 ± 13.2 years and mean follow-up following treatment was 37.2 ± 31.8 months. On average, amputated patients gained 0.018 ± 0.386 points on the EQ-5D-3 L index, whereas those treated by external fixation exhibited an improvement of 0.624 ± 0.229 points ($p = 0.005$).

Incremental costs

The incremental costs associated with each treatment are summarized in Table 3. The cost of surgery is logically higher in the case of external fixation as it includes the cost of the fixator itself and of the operation required to remove it (€17,779 vs. €35,272; $p < 0.001$). The costs of the subsequent follow-up of patients are similar for both groups (€60,704 vs. €66,519; $p = 0.465$).

Health outcomes

The QALYs obtained with each treatment option, as calculated according to the model, are included in Table 4 and represented in Fig. 3. Consequently, the mean QALE for an amputated patient over a 10-year period stood at 2.641 ± 1.64 QALYs, while that for a patient treated with external fixation was 6.627 ± 1.46 QALYs ($p < 0.001$).

Cost-utility

The mean CER stood at €29,717/QALY for amputees, and at €15,360/QALY for patients undergoing external fixation.

Table 3 Incremental costs

	Amputation	External fixation	P value
Surgery related costs	€17,902	€35,741	< 0.001*
Follow up costs			
Pharma	€353	€1,347	0.006
Transport and hospital stay	€40,236	€61,236	0.984
Other surgeries	€1,075	€1,933	0.443
Rehabilitation	€18,917	€1,534	< 0.001*
Mean total surgery related costs	€17,902	€35,741	< 0.001*
Mean total follow up costs	€60,581	€66,050	0.465
Mean total incremental costs	€78,483	€101,791	0.767

* Statistically significant differences between the groups

Table 4 Health outcomes (quality of life measured using the EQ-5D index)

Year	External fixation (QALYs/year)	Amputation (QALYs/year)
1	0.500	0.507
2	0.776	0.421
3	0.741	0.376
4	0.724	0.303
5	0.707	0.241
6	0.680	0.196
7	0.649	0.176
8	0.640	0.159
9	0.612	0.145
10	0.598	0.118
Total (QALE)	6.627	2.642

$$CEM_{AMP} = \frac{€78,483}{2.641 \text{ QALYs}} \approx €29,717/\text{QALY}$$

$$CEM_{FE} = \frac{€101,791}{6.627 \text{ QALYs}} \approx €15,360/\text{QALY}$$

And the comparison between both options resulted in an ICER of €5,847/QALY.

$$CEI = \frac{\Delta C}{\Delta R} = \frac{€101,791 - €78,483}{6.627 \text{ QALYs} - 2.641 \text{ QALYs}} \approx €5,847/\text{QALY}$$

To determine whether this value was cost-effective, it was compared with the main national and international references and was found to be below all published effectiveness thresholds [50, 51, 64–67]. Figure 4 represents the cost effectiveness of the interventions.

Sensitivity of the cost-utility analysis

Table 5 summarizes the results of the sensitivity analyses performed. ICER stood in all cases below the effectiveness thresholds. Even in the most unfavorable scenario (which includes higher fixator costs, poorer outcomes of external fixation and better results for amputation) the ICER remained at €12,296/QALY.

Budgetary impact analysis

Table 6 presents a 10-year budgetary impact analysis, apart from the sensitivity analyses conducted with respect to it. Although from a statistical point of view the cost of neither treatment is higher than that of the other, as a result of the longer life expectancy enjoyed by patients treated with external fixation and of the relatively higher cost of this therapeutic option, the budgetary impact analysis revealed that a greater economic effort was required to pay for the external fixator, which entailed a mean cost of €52.8 million per year.

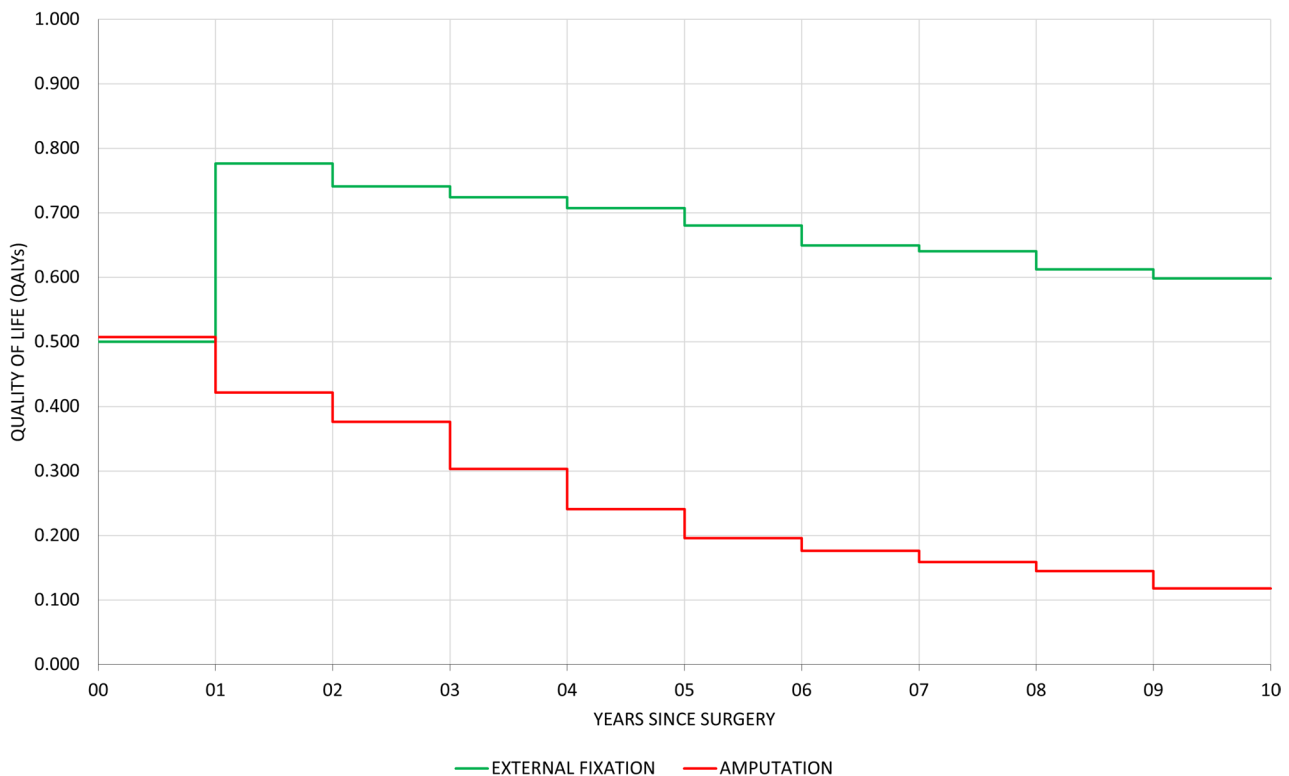


Fig. 3 Health outcomes predicted by the model. The red line indicates the health outcomes for the amputation group, while the green line reflects those for the external fixation group. The areas under the curves represent the quality-adjusted life expectancy, while the difference between them is a reflection of the incremental health outcomes

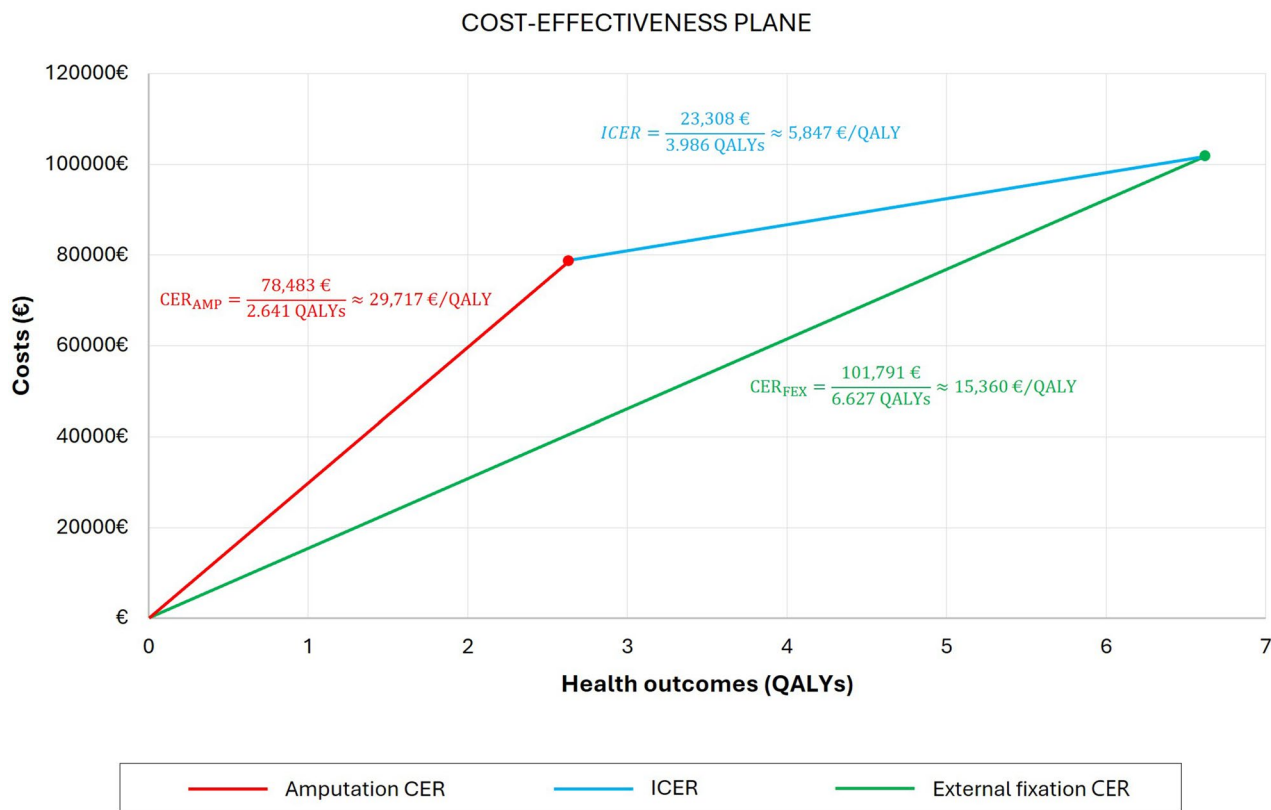


Fig. 4 Cost-effectiveness representation showing the mean cost-effectiveness ratio (CER) for each treatment alternative and their incremental cost-effectiveness ratio (ICER)

Table 5 Sensitivity analysis on the cost-utility analysis

Analysis	Modification of parameters	QALE (QALYs)		Incremental costs		ICER
		Amputation	Ext. Fix.	Amputation	Ext. Fix.	
-	Standard model	2.641	6.627	€78,483	€101,791	€5,847/QALY
SA-1	Cost of ext. fixator €20,000 higher	2.641	6.627	€78,483	€121,791	€10,865/QALY
SA-2	Poorer clinical outcome for external fixation	2.641	6.295	€78,483	€101,791	€6,379/QALY
SA-3	Better clinical outcome for amputation	2.773	6.627	€78,483	€101,791	€6,048/QALY
SA-4	SA-2 + SA-3 + SA-4	2.773	6.295	€78,483	€121,791	€12,296/QALY

Discussion

A previous study demonstrated that, with similar costs, external fixation results in a greater improvement in quality of life than amputation in the management of the ulcerated diabetic foot [37]. In addition, the present study confirms that external fixation is a cost-effective alternative.

Interpretation of health outcomes

A cost-utility analysis does not permit acceptance or rejection of a treatment alternative in absolute terms; a comparison must be made with another intervention used as a comparator [62]. Although the mean CER value may be considered informative, it cannot be used as a deciding criterion by itself. Moreover, interventions with a higher CER are usually less cost-effective [63].

Our analysis estimates that amputation is associated with a mean CER of €29,717/QALY. Given that mean CER is equivalent to the ICER when an intervention is compared with “doing nothing”, an ICER of the said value would be at the limit of being considered cost-effective according to most effectiveness thresholds. Nonetheless, exceptions such as the criteria applicable to end-of-life situations must be considered as these could increase the cost-effectiveness threshold to as much as £50,000/QALY [51]. These thresholds could be considered applicable to patients with ulcerated diabetic foot as the absence of treatment would result in the patient’s death. In any case, the authors are not aware of any study evaluating the cost-effectiveness of amputation in these cases.

Treatment by external fixation provides better health outcomes than amputation (2.641 QALYs in 10 years vs.

Table 6 Budgetary impact analysis

Analysis	Treatment	Detail	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	TOTAL
Standard	Amputation	Yearly surgeries	2,835	2,721	2,612	2,508	2,407	2,311	2,219	2,130	2,045	1,963	23,751
		Survivors	2,436	4,479	6,251	7,785	9,147	10,208	11,078	11,853	11,376	10,922	-
		Cost during 1st year	50.8 M	48.7 M	46.8 M	44.9 M	43.1 M	41.4 M	39.7 M	38.1 M	36.6 M	35.1 M	425.2 M
		Follow-up expenses	73.8 M	135.7 M	130.2 M	125.0 M	120.0 M	115.2 M	110.6 M	106.2 M	101.9 M	97.9 M	1116.6 M
		Total	124.5 M	184.4 M	177.0 M	169.9 M	163.1 M	156.6 M	150.3 M	144.3 M	138.5 M	133.0 M	1541.8 M
	Ext. Fix.	Updated total	120.9 M	173.8 M	162.0 M	151.0 M	140.7 M	131.1 M	122.2 M	113.9 M	106.2 M	99.0 M	1320.8 M
		Yearly surgeries	2,835	2,721	2,612	2,508	2,407	2,311	2,219	2,130	2,045	1,963	23,751
		Survivors	2,730	5,263	7,635	9,855	11,948	13,857	15,629	17,301	16,609	15,943	-
		Cost during 1st year	101.3 M	97.3 M	93.4 M	89.6 M	86.0 M	82.6 M	79.3 M	76.1 M	73.1 M	70.2 M	848.9 M
		Follow-up expenses	82.7 M	159.4 M	153.0 M	146.9 M	141.0 M	135.4 M	130.0 M	124.8 M	119.8 M	115.0 M	1308.0 M
SA-6 Constant number of surgeries	Amputation	Total	184.0 M	256.7 M	246.4 M	236.5 M	227.1 M	218.0 M	209.3 M	200.9 M	192.9 M	185.2 M	2156.9 M
		Updated total	178.7 M	241.9 M	225.5 M	210.2 M	195.9 M	182.5 M	170.2 M	158.6 M	147.8 M	137.8 M	1849.0 M
		Updated BALANCE [ext. fix. - amputation]	57.7 M	68.1 M	63.5 M	59.2 M	55.2 M	51.4 M	47.9 M	44.7 M	41.7 M	38.8 M	528.2 M
		Amputation - Updated total	120.9 M	178.5 M	173.3 M	168.3 M	163.4 M	158.6 M	154.0 M	149.5 M	145.2 M	140.9 M	1552.6 M
		Ext. Fix. - Updated total	178.7 M	248.9 M	241.7 M	234.6 M	227.8 M	221.2 M	214.7 M	208.5 M	202.4 M	196.5 M	2174.9 M
	SA-7 5% cost discount	Updated BALANCE [ext. fix. - amputation]	57.7 M	70.4 M	68.3 M	66.4 M	64.4 M	62.5 M	60.7 M	59.0 M	57.2 M	55.6 M	622.3 M
		Amputation - Updated total	118.6 M	167.2 M	152.9 M	139.8 M	127.8 M	116.9 M	106.8 M	97.7 M	89.3 M	81.7 M	1198.7 M
		Ext. Fix. - Updated total	175.3 M	232.8 M	212.8 M	194.6 M	177.9 M	162.6 M	148.7 M	136.0 M	124.3 M	113.7 M	1678.8 M
		Updated BALANCE [est. fix. - amputation]	56.6 M	65.6 M	60.0 M	54.8 M	50.1 M	45.8 M	41.9 M	38.3 M	35.0 M	32.0 M	480.1 M

6.627 QALYs; $p < 0.001$) and, given that the incremental costs of both procedures are similar ($p = 0.767$), it comes as no surprise that the ICER demonstrates the cost-effectiveness of this treatment option (between €5,847/QALY and €12,296/QALY, according to the sensitivity analysis performed). Furthermore, external fixation is cost-effective across all the different scenarios tested, including SA-4 (the worst scenario for external fixation), where the clinical outcomes of amputation are overstated, the outcomes of external fixation are underestimated, and the cost of the external fixator is artificially increased. It must be noted that, although the cost of the external fixator is often a controversial aspect in the treatment, it does not have a significant impact on its ICER.

Interpretation of the budgetary impact

Several factors may be invoked to explain the differences in the estimated budget for each treatment option. The most significant one is probably the longer survival observed among non-amputated patients. A 10-year extrapolation of the data included in the study used as a basis for the current analysis [37] estimated that only 35.4% of amputated patients would survive over that period, while 77.2% of those treated with external fixation would stay alive. This situation, which has highly positive effects from a social standpoint (at the end of the period there would be a difference between the two treatment alternatives of 5,021 survivors), has a significant effect on the budget as costs would significantly increase.

On the other hand, differences also exist with respect to cost structures. Although the difference between the incremental costs of both treatments has not been shown to be statistically significant, the costs related to the procedure itself are slightly higher for external fixation, which results in higher costs during the first year post-op.

Rationale for the model

In the current socioeconomic context where optimization of healthcare costs is fundamental, cost-effectiveness analyses have become a key decision-making tool for funding healthcare technologies. As a result, many countries have established agencies specializing in evaluating the relationship between costs and health benefits, or have incorporated the requirement to establish such bodies in their legislation.

An analysis of this nature requires an evaluation of the economic costs and the health outcomes associated to one treatment and its comparator. To the best of the authors' knowledge, apart from the study used as a basis for the present economic analysis [37], there is only one other study that considers the costs associated with the management of diabetic foot by means of external fixation [22], and no studies that evaluate health outcomes by means of QALYs.

However, although the length of follow-up in the study used as a basis for this analysis [37] is too short to carry out a cost-utility evaluation, modeling and simulation are valid techniques for this kind of investigation [38, 70]. The present study used a discrete-element model that makes it possible to assign individualized quality of life and utility values to each patient [71, 72], analyzing their evolution over time and limiting them on occurrence of discrete events such as a patient's death [72, 73], the probability of which is modeled based on the probabilities obtained in the study that served as a basis for this analysis.

Clinical and economic implications

According to the American Limb Preservation Society, *once an amputation is performed, the next step on the express lane is often death* [74]. With mortality rates in excess of those of many tumors [4–7] and with costs equivalent to those associated with the treatment of cancer [12], it seems evident that all research efforts should be focused on developing effective alternatives able to reduce the impact that disease has on health as well as the economic burden it places on the system.

In line with other authors, the study serving as a basis for this analysis found that external fixation led to superior results in terms of clinical outcomes and survival than amputation [31, 32, 34, 35, 37], in addition to reporting similar costs for both treatment alternatives [37]. The present economic analysis also confirms that external fixation is a cost-effective alternative as measured against the usual cost-effectiveness thresholds and that budgetary increases are not as high as to be considered unjustified, particularly considering that they typically result from an increase in the percentage of survivors.

Moreover, there are some costs which – although not considered in the present analysis – must be taken into account by decision-makers, such as dependent care and palliative care expenses. As regards the former, we have found no studies evaluating the economic impact of an amputated diabetic patient for social welfare systems. As far as the latter are concerned, a case-control study on 216,413 diabetic patients with peripheral arterial disease demonstrated that amputees were less likely to die at home, stayed in hospital twice as long as non-amputees, and generated nearly twice as many expenses in their last 90 days of life [75].

Limitations and strengths of the study

In the first place, the population included in the study that served as a basis for this analysis was not obtained from a randomized clinical trial, but from a before-and-after study, which is warranted in cases where the scientific community considers that a new treatment is associated with better outcomes than a previous one [76].

We therefore consider it highly unlikely that an ethics committee may in the future approve a randomized clinical trial comparing the two treatment options discussed here. In addition, although clinical trials are associated with a high degree of internal validity, their applicability to the real world is limited by their failure to address externalities, the inclusion of unrealistic costs, and the fact that patients are rarely followed up in a normal clinical situation [77]. In the study serving as a basis for the present one [37], costs were estimated by micro-costing, using market prices and in the context of routine clinical practice, which maximizes its external validity [42]. In this regard, it is worth pointing out that, according to some authors, in the absence of randomized clinical trials, it is acceptable to resort to literature sources with lower levels of evidence provided that their use is duly warranted [43].

A more significant limitation is the small size of the sample used, which may affect the consistency of the results. However, the size is comparable to that of similar previously published studies and its findings are consistent with theirs [31, 32, 34].

Lastly, it must be said that the model used is overly simple as it does not include data on other variables of interest such as sex, comorbidities or obesity which typically influence patient mortality. For that reason, the only possible predictors are the results of the with treatment alternatives compared and the life expectancy recorded for diabetic patients. Nonetheless, it should be remembered that the purpose of an economic evaluation is not to provide an accurate prediction of the outcomes or the costs of a treatment, but rather to provide information that can be used for decision-making at a given time [45]. Moreover, the estimate presented here is based on the only clinical trial comparing two homogeneous patient cohorts, where all the costs associated with the treatment of each patient were recorded exhaustively and in routine clinical practice conditions. At the same time, the relationship that exists between the clinical outcomes of the treatment alternatives analyzed makes it highly unlikely for *clinical equipoise* to warrant the conduct of similar prospective clinical studies in the future.

Conclusions

The use of an external fixation in the treatment of diabetic foot ulcers has been shown to be a cost-effective option as compared with major amputations.

Abbreviations

ICER	Incremental cost-effectiveness ratio
QALYs	Quality-adjusted life years
QALE	Quality-adjusted life expectancy
CER	Cost-effectiveness ratio

Author contributions

P.O.R.M. participated in the conception and design of the work, the analysis and interpretation of data, and the drafting and revision of the manuscript.L.L.C. participated in the conception and design of the work, the collection and interpretation of data, and the revision of the manuscript.M.R.R. participated in the conception and design of the work, the interpretation of data, and the revision of the manuscript.A.S.F. participated in the conception and design of the work, the collection and interpretation of data, and the revision of the manuscript.N.G.R. participated in the conception and design of the work, the interpretation of data, and the revision of the manuscript.E.G.F. participated in the conception and design of the work, the interpretation of data and the revision of the manuscript.All authors approved the submitted version and are responsible for the accuracy and integrity of the work.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of the Consorci Sanitari Integral Hospital (code CEIC HUB PR367/21) and all patients gave their informed consent before inclusion.

Consent for publication

Not applicable.

Competing interests

M.R.R. and P.O.R.M. are affiliated with the MBA Institute and the MBA Institute Chair for Clinical and Biomechanical Research of the University of Oviedo. The rest of the authors have no competing interests to declare.

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